

Local AI through the browser

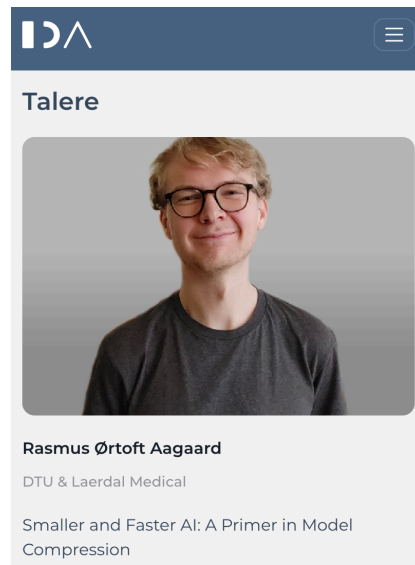
Wait, it's just a bit of Javascript?

github.com/rasgaard/slides

Who am I?



- Rasmus Aagaard
- Industrial PhD Student
 - Laerdal Medical & Technical University of Denmark
 - Research on model compression and on-device deployment
- Data Scientist at Laerdal Medical for 1,5 years
 - LLM integrations, evals and internal tooling
- Likes to read, drink beer and climb boulders



Also at IDA Driving AI (ida.dk/driving-ai) for more yapping about small on-device models

What is *Laerdal Medical* and what does it do?

- *"Our goal is helping save one million more lives. Every year. By 2030."*
- Manikins for medical training
- Resuscitation training
- In Copenhagen, digital simulation
 - Virtual/Mixed reality scenarios
 - *Video games for nursing students*

You have been blocked from The New York Times because we suspect that you're a robot.

Why am I seeing this? There are a few possibilities:

- You are browsing and clicking much faster than is typical of a human being
- Something is preventing Javascript from working on your computer
- There is a bot with the same IP address as you

Having trouble? [Contact support.](#)

ID: 140d54af-ddee-2b80-20b2-f39df1fd41d3

The real claim to fame

Motivation

Why this meetup? A little side project

Motivation



Kasper Junge • 1st
AI Engineer @ Dinero 🔥 Underviser Dev Teams i Vibe Coding 🧠 LLMs a...
2w • 🌐

Rasmus Aagaard har bygget en open source web-baseret transskribering service med Whisper-large-v3-turbo og Transformers.js som du kan prøve her: <https://lnkd.in/eGuPNvxK>

Det hele kører i browseren!

Tjek koden her: <https://lnkd.in/eiRBPEEU>

Vær med i diskussionen på på Rasmus post på [Danish Data Science Community](#) slack her:
<https://lnkd.in/eZMKzTRW>

Hvis du ikke allerede er medlem af DDSC slack kan du blive det her:
<https://lnkd.in/dDM7YkmT>

Show translation

AUDIO TRANSCRIPTION SYSTEM V2.0 [READY]

LANGUAGE:
ENGLISH ▾

AUDIO SOURCE:
[CLICK TO SELECT FILE]

TRANSCRIBE

[COPY OUTPUT]

TIMESTAMP	TRANSCRIPTION DATA
[NO AUDIO FILE LOADED]	

 You and 71 others

9 comments · 1 repost

Motivation

 **Mike Riess** ✓ • 1st 2w ...
PhD., Senior Research Scientist - AI @ Telenor R&I

Rasmus Agaard øhhh kører det hele i de to js filer? 🤔

Show translation

Like · ❤️ 1 | Reply

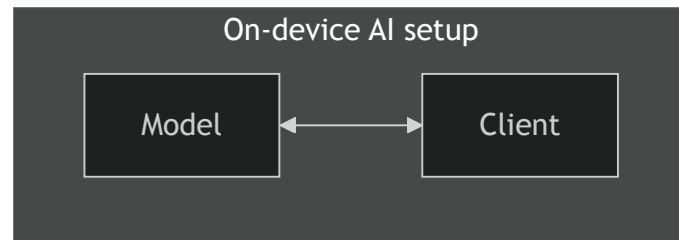
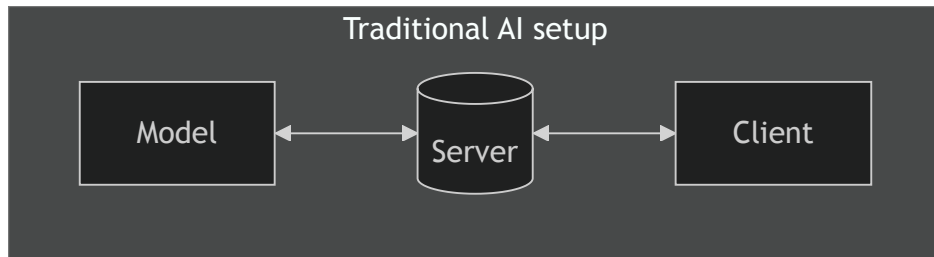
 **Jonas Akrouh Larsen** ✓ • 2nd 4d ...
CIO / Azure / Digitalisation / Modern workplace

Jeg forstår det simpelthen ikke 😞 Kører den selve modellen som javascript i klientens browser? Det lyder helt crazy? Er det så ren CPU?

Show translation

Like | Reply · 2 replies

Cause of confusion



We're used to dealing with a bunch of complications out of necessity

Common AI deployment headaches?

- ...

How do deploy on-device through the browser?

"Just go to this website"



YOUR AUDIO NEVER LEAVES YOUR DEVICE

All transcription is done locally on your computer. Your audio files are processed entirely in your browser - nothing is uploaded to any server.

LANGUAGE:

ENGLISH ▼

AUDIO SOURCE:

[CLICK TO SELECT FILE]

TRANSCRIBE

[COPY OUTPUT]

TIMESTAMP

TRANSCRIPTION DATA

[NO AUDIO FILE LOADED]

Q: How is this possible? A: Transformers.js

```
from transformers import pipeline

# Allocate a pipeline for sentiment-analysis
pipe = pipeline('sentiment-analysis')

out = pipe('I love transformers!')
# [{'label': 'POSITIVE', 'score': 0.999806941}]
```

Python

```
import { pipeline } from '@huggingface/transformers';

// Allocate a pipeline for sentiment-analysis
const pipe = await pipeline('sentiment-analysis');

const out = await pipe('I love transformers!');
// [{'label': 'POSITIVE', 'score': 0.999817686}]
```

Javascript

Core of transcription site

<https://github.com/rasgaard/webai-stuff/tree/main/transcribe>

```
import { pipeline } from '@huggingface/transformers';

const transcriber = pipeline("automatic-speech-recognition",
  "onnx-community/whisper-large-v3-turbo", {
    dtype: {
      encoder_model: "fp16",
      decoder_model_merged: "q4"
    },
    device: "webgpu",
  });
```

```
const output = await transcriber(audio, {
  return_timestamps: true
});
```

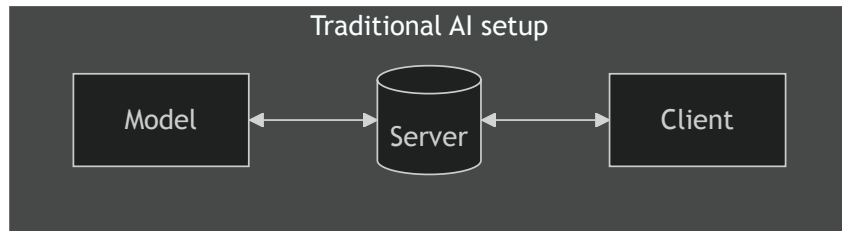
Trade-offs?

Pros

- You're in full control
- Data stays on the device
- Azure is down? Not a problem

Cons

- Lose some flexibility
- Large initial downloads



More demos

How to get started?

Follow the lead developer, Joshua Lochner (Xenova)

- <https://www.linkedin.com/in/xenova/recent-activity/all/>

Take a look at the demos

- <https://huggingface.co/collections/webml-community/transformersjs-v4-demos>

and take time to really *read (**study**) the code*

- <https://github.com/huggingface/transformers.js-examples>

Finally, think of use-cases and implement those. Claude is a good helper :)

Thanks for listening!

Let's continue the conversation :)

DDSC Slack
#on-device-ai



Connect on LinkedIn

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